

Replication of The Valuation of Exotic Options Using SPD with Higher Moments

Stochastic Processes in Financial Engineering Project

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1 Introduction

In financial economics, the valuation of derivative securities is fundamentally grounded in the principle of no-arbitrage. Under this framework, the price of a derivative can be expressed as the discounted expected value of its future payoff under an appropriate state-price(SPD)(Cox and Ross, 1976). Formally, the no-arbitrage price of a derivative is given by:

$$\text{Price} = e^{-rT} \int G(x) h(x) dx$$

where $G(x)$ denotes the payoff function and $h(x)$ represents the state-price density of the underlying asset at maturity. In standard settings, such as the Black-Scholes framework, the SPD is lognormal, allowing for closed-form solutions for plain vanilla options Scholes and Merton, 1973.

However, this analytical tractability breaks down when dealing with exotic derivatives. Instruments such as Asian, basket, and spread options depend on the underlying stochastic process that deviates from log-normality, making it difficult to derive closed-form expressions for their state-price densities. As a result, pricing these derivatives often relies on computationally intensive methods such as Monte Carlo simulation or numerical approximation techniques (Boyle et al., 1989; Turnbull and Wakeman, 1991).

Although several alternative approaches have been proposed, these methods typically involve trade-offs between computational efficiency and pricing accuracy. In particular, many approximation techniques struggle to capture higher-order distributional characteristics, such as skewness and kurtosis, which are essential for accurately valuing exotic derivatives(Jarrow and Rudd, 1982; Corrado and Su, 1997).

This study aims to address these limitations by adopting a flexible approximation of the state-price density based on moment-matching techniques. In particular, the approach builds on the use of the Johnson(1949) family of distributions, which provides a flexible framework capable of incorporating higher moments of the underlying distribution. By doing so, the method seeks to retain analytical tractability while improving the pricing accuracy for a wide class of exotic options.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on exotic option pricing and approximation techniques. Section 3 outlines the methodology and presents the proposed state-price density approximation framework. Section 4 reports the empirical results, while Section 5 concludes the paper. Section 6 is an appendix.

2 Literature Review

2.1 Early Option and Warrant Pricing

Financial derivative pricing is based on a simple principle: correctly priced derivatives should not allow for arbitrage opportunities through long or short positions on the options or underlying assets (Black and Scholes, 1973). Early work on the valuation of options before the development of the Black-Scholes model, options were predominantly valued in terms of warrants. Warrants give holders the right but not the obligation to purchase a company's shares at a predetermined price before an expiry date (Chen, 1970). This is very similar to a call option as both these instruments allow the holder to purchase some underlying stock at a fixed strike price and are both exercised only when market price of the underlying is greater than the strike.

Early attempts at warrant pricing, such as by Sprenkle (1961), valued underlying asset prices as lognormally distributed and derived expected value pricing formulas, which built the foundation for later developments. The lognormal distribution assumes that the asset prices are always positive and continuously compounded return follow a normal distribution (Hull, 2018). Sprenkle's option valuation formula closely resembles the Black-Scholes formula and thus suggests that researchers were close to the modern option formula structure. In the formulation, the value of the option is shown a component of the underlying stock price and the exercise price, both which are weighted by a normal distribution (Sprenkle, 1961). This formulation can be written as:

$$\begin{aligned} W &= kx N(b_1) - kc N(b_2) \\ b_1 &= \frac{\ln\left(\frac{kx}{c}\right) + \frac{1}{2}v^2(t^* - t)}{v\sqrt{t^* - t}} \\ b_2 &= \frac{\ln\left(\frac{kx}{c}\right) - \frac{1}{2}v^2(t^* - t)}{v\sqrt{t^* - t}} \end{aligned} \tag{2.1}$$

where:

- X is the stock price,
- C is the strike price,
- T^* is the maturity date,
- T is the current date,
- V^2 is the variance of stock returns,

- $N(b)$ is the cumulative standard normal distribution function.

This formula already recognised that an option's value is dependent on the current price, strike price, volatility, and time to maturity. However, the model was incomplete as the formula contained unknown parameters k and k^* . Sprenkle (1961) defines these parameters as follows: k represents the ratio of expected future stock price to the current stock price, while k^* represents the discount factor and is dependent on the riskiness of the stock. The main limitation of this formulation was that these parameters had to be estimated empirically, which Sprenkle failed to do. Therefore, although the formula structurally resembled the Black–Scholes–Merton (BSM) framework, it did not provide a complete option-pricing theory.

Samuelson (1965) also used the expected-value approach with its assumptions; however, his model still relied on two unknown parameters: α , the expected rate of return on the stock, and β , the discount return rate applied to the warrant. This model valued the warrant by taking the expected value of the future stock-price distribution above the exercise price and then discounting the value back to the present at rate β . While this captured the idea of a warrant's value being dependent on its expected payoff when exercised, it was still incomplete as there was no capital market model adequate to determine the appropriate discount rate (Samuelson, 1965).

This weakness was recognised by Samuelson and Merton (1969), who argued that discounting the expected payoff was not the appropriate valuation method. They therefore improved the framework by treating the option price as a function of the underlying stock price and recognising that the appropriate discount rate must reflect a portion of systematic risk. While this was a conceptual advance, the formula remained dependent on investor risk preferences, and no preference-free solution could be obtained.

2.2 Dilution and Expected Payoff Framework

Extending on the probabilistic foundation established by Sprenkle (1965) and others, Chen (1970) developed a structurally more complete framework for warrant pricing. Chen highlighted that the warrant's value depends on whether the stock price at time T is above the strike price. And the expected value of the warrant is:

$$EVW_t = \int_{y_t}^T (x_t - y_t) f(x_t) dx_t \quad (t \leq T) \quad (2.2)$$

$$EVW_t = 0 \quad (t > T)$$

where:

- EVW_t is the expected value of the warrant t periods from now,
- x_t is the market price of the associated stock at time t ,
- y_t is the exercise price at time t ,
- $f(x_t)$ is the probability density function of x_t ,
- T is the expiration date of the warrant.

The formula states that a rational warrant holder only exercises if the market price x_t is greater than the exercise price y_t , in which case the payoff becomes $x_t - y_t$. If the price is lower, the warrant is not exercised. Therefore, the integral is evaluated from y_t to infinity, giving the sum of all possible outcomes weighted by their probability of occurrence under the density function $f(x_t)$. After the expiry date, the warrant has no optionality and its value is zero.

This equation represents the expected payoff of a call option evaluated under the probability density function $f(x_t)$ of the underlying asset price:

$$E[(X_t - y_t)^+].$$

Importantly, the formulation is general with respect to $f(x_t)$ and does not impose a specific distributional assumption on the stock price. Chen, however, specifies $f(x_t)$ as lognormal, making the model consistent with Sprenkle (1961) and effectively yielding a closed-form solution. Under the lognormal assumption, the integral can be evaluated analytically, producing a formula structurally similar to Sprenkle's but extended to account for a unique feature of warrants.

This feature is the dilution of existing shareholder value (Galai and Schneller, 1978). When warrants are exercised, new shares are issued by the company, increasing the total number of outstanding shares and thereby reducing the value of existing shares on a per-share basis. Given a firm with n shares outstanding and m warrants exercisable at strike price K , the post-exercise share price becomes:

$$S^* = \frac{nS + mK}{n + m} \tag{2.3}$$

The payoff therefore becomes:

$$\max(S^* - K, 0)$$

rather than:

$$\max(S - K, 0).$$

Any model that ignores this characteristic overestimates the value of the warrant (Galai and Schneller, 1978). This dilution adjustment therefore represents a more economically meaningful warrant pricing framework, as it produces valuations that are realistic when accounting for warrant exercise by scaling the effective share value through the factor $\frac{n}{n+m}$.

Nevertheless, this framework remains dependent on the expected rate of return of the underlying asset, a parameter that is investor-specific and unobservable. As a result, the model remains incomplete as a market-based pricing tool. This limitation was later resolved by Black and Scholes (1973) and Merton (1973) through the introduction of no-arbitrage replication techniques.

2.3 The Black-Scholes-Merton Framework

Black and Scholes (1973) and Merton (1973) resolved the central limitation of all prior warrant and option pricing models through a replication argument. Rather than pricing an option as the expected payoff under subjective beliefs, as aforementioned, they demonstrated that under continuous trading and no transaction costs, a portfolio of the underlying asset and a risk-free bond can be continuously rebalanced to perfectly replicate the option's payoff at every point in time. Since this replicated portfolio produces similar cashflows to the option, the no-arbitrage condition states that the option price must equal the cost of constructing that portfolio, regardless of investor preference. Therefore, the expected return of the underlying is removed entirely from the equation. Black and Scholes (1973) indicated that using the assumption that the underlying asset follows a geometric Brownian motion, the no-arbitrage conditions implies that the option price satisfies a partial differential equation whose solution for a European call option is:

$$C = Se^{-rT}N(d_1) - Ke^{-rT}N(d_2) \quad (2.4)$$

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

where:

- S is the current asset price,
- K is the strike price,
- T is the time to maturity,

- r is the risk-free rate,
- q is the dividend yield,
- σ is the volatility of the underlying asset,
- $N(\cdot)$ is the cumulative standard normal distribution function.

Every input in the model is either directly observable or specified within the option contract. Comparing Equation (1.4) with Equation (1.1) shows the precise resolution: under risk neutrality, Sprenkle's k becomes e^{rT} and k^* becomes 1, both of which are fully determined by observable market parameters.

This framework was further extended by Merton (1973), who derived a continuous-time formulation using Itô's lemma and incorporated dividend-paying assets. Together, the Black–Scholes–Merton (BSM) framework advanced derivative pricing from a problem of subjective expectations to an objective, market-consistent replication approach.

However, its elegance is dependent on the assumption that the underlying asset follows a lognormal distribution (Black and Scholes, 1973; Hull, 2018). While this assumption is widely accepted and tractable for plain European options on a single asset, it becomes restrictive in the context of exotic options. As a result, the BSM framework cannot be generalised to all option types.

2.4 State-Price Densities

Arrow (1964) and Debreu (1954) introduced the concept of “state-contingent claims”, hypothetical assets that deliver one unit of a reference asset if a particular state of the world occurs, and zero otherwise. In complete markets, the price of an asset can be viewed as a weighted combination of these elementary claims. The weights, known as state prices or Arrow–Debreu prices, capture the market's valuation of each possible future state.

Harrison and Kreps (1979) extended no-arbitrage pricing to continuous time by showing that derivative prices can be valued under a risk-neutral probability measure. Under this measure, discounted asset prices behave as martingales and are consistent with the absence of arbitrage. In continuous-time finance, this leads to the concept of a state-price density (SPD), also known as the risk-neutral density (Debreu, 1954).

This SPD assigns a probability weight to every possible terminal asset price under the risk-neutral measure. Under this formulation, the no-arbitrage price of any derivative is the discounted expectation of its payoff under the risk-neutral measure. Posner and Milevsky (1998) formalised this result as:

$$\text{Price} = e^{-rT} \int_0^\infty G(x) h(x) dx \quad (2.5)$$

where:

- $G(x)$ denotes the payoff function,
- $h(x)$ represents the state-price density.

From this perspective, the Black–Scholes–Merton model can be interpreted as the discounted expected payoff of an option evaluated by integrating against a lognormal SPD. The Arrow–Debreu pricing framework therefore provides the rigorous theoretical basis for state-price density approximation methods. The pricing problem for any exotic option reduces to identifying a suitable $h(x)$, whether exact or approximate, and integrating the payoff function against it.

2.5 Exotic Options

While the above framework provides a general pricing principle, it is limited by a fundamental distributional problem. Asian options are options whose payoffs are based on the arithmetic average of the asset’s price over a specified time period, while basket options have payoffs dependent on multiple underlying assets (Song, 2024). Lévy (1992) showed that such aggregates are not lognormally distributed, implying that the Black–Scholes–Merton (BSM) stock price distribution is inapplicable and that the pricing integral does not admit a closed-form solution.

In the absence of closed-form solutions, the distributional properties of exotic options are often characterised through their statistical moments. Let X^T denote the stochastic element of the payoff and $F(X^T)$ its risk-neutral expectation. As per Posner and Milevsky (1998), the first four central moments are the mean, variance, skewness, and kurtosis, and the generalised forward price of X^T is defined as:

$$F(X^T) = \mathbb{E}[X^T] = \int_0^\infty x h(x) dx.$$

The dispersion and higher-order shape characteristics of the distribution are defined as follows. The standard deviation is given by:

$$\zeta = \sqrt{\mathbb{E}[(X^T - F(X^T))^2]},$$

the skewness is defined as:

$$\eta = \frac{\mathbb{E}[(X^T - F(X^T))^3]}{\zeta^3},$$

and the kurtosis is given by:

$$\kappa = \frac{\mathbb{E}[(X^T - F(X^T))^4]}{\zeta^4}.$$

Importantly, these moments can be computed analytically even when the full distribution is intractable (Turnbull and Wakeman, 1991). This supports the moment-matching approach: once the first four moments of the true state-price density are known, they can be used to fit a simpler auxiliary distribution that permits closed-form pricing.

2.6 Benchmark Methods

In the absence of closed-form solutions, exotic options are commonly priced using numerical methods, particularly Monte Carlo simulation, which estimates value by averaging discounted simulated payoffs and is widely used as an accuracy benchmark (Boyle, 1977). Other approaches include multi-asset lattice methods, geometric conditioning approximations for Asian options, and analytical lower bounds; however, these methods often involve trade-offs between accuracy, computational speed, and scalability (Rogers and Shi, 1995).

Although these techniques can be highly accurate, they are often computationally expensive in the context of risk management for large portfolios of exotic options. This has therefore motivated the use of analytical approximation methods, which trade small losses in accuracy for significantly faster pricing and efficient repeated valuation.

2.7 Moment-Matching Approximations

The most natural analytical approach to the state-price density (SPD) approximation problem is to replace the unknown true density $h(x)$ with a tractable approximating density calibrated to match the known moments of the exotic payoff. Jarrow and Rudd (1982) used Edgeworth expansions to approximate the unknown payoff distribution by expanding around a simpler reference density and matching statistical moments. Essentially, their framework replaces the intractable true SPD with a tractable proxy that matches its lower-order moments and exploits the proxy's analytical properties to obtain a closed-form price. This framework is the direct conceptual ancestor of the four-moment method of Posner and Milevsky (1998).

Turnbull and Wakeman (1991) applied Edgeworth expansions to arithmetic Asian options

by matching the first four cumulants of the arithmetic average, improving on Lévy's (1992) two-moment lognormal approximation. However, this approach remains linked to the lognormal structure and becomes less reliable for high-volatility assets. Corrado and Su (1997) extended the Jarrow–Rudd approach and showed that skewness and kurtosis can be inferred from index option prices. Their results demonstrate that higher-order moments are critical in market pricing and further support the use of moment-matching methods.

The main limitation of this framework is that Edgeworth expansions can produce negative density values in the tails and may violate the fundamental requirement of a valid probability distribution. Furthermore, the approach does not fully replace the lognormal distribution; rather, it only corrects it. As a result, it is most reliable when the true distribution is close to lognormality (Turnbull and Wakeman, 1991). This motivates the use of alternative distributions that can match moments while remaining non-negative over the entire support.

2.8 The Johnson Distribution Family

To address the limitations of the Edgeworth expansion, the choice of approximating distribution is critical. Johnson (1949) introduced a system of statistical distributions defined by four parameters, corresponding to the first four moments. The Johnson family provides a more complete moment-matching framework, as it can reproduce mean, variance, skewness, and kurtosis while always generating a valid non-negative probability density function (Johnson, 1949). It includes both bounded and unbounded forms, allowing flexibility in tail behaviour while still permitting closed-form integration of payoffs.

The Johnson system generates distributions through a smooth transformation of a standard normal variable Z , and is generally defined as:

$$X = c + d\phi^{-1}\left(\frac{Z - a}{b}\right)$$

The choice of transformation $\phi(\cdot)$ determines the distributional form. If $\phi(\cdot) = \ln(\cdot)$, the resulting distribution is the shifted lognormal (Type I, S_L), given by:

$$\text{Type I } (S_L): \quad X = c + d\exp\left(\frac{Z - a}{b}\right)$$

When $\phi(\cdot) = \sinh^{-1}(\cdot)$, the distribution is unbounded (Type II, S_U), given by:

$$\text{Type II } (S_U) : \quad X = c + d \sinh\left(\frac{Z - a}{b}\right)$$

where a , b , c , and d are free parameters and $Z \sim \mathcal{N}(0, 1)$. Type I generalises the Black–Scholes–Merton lognormal model through the location parameter c , which shifts the distribution, and the scale parameter d , which stretches or compresses it. The BSM lognormal distribution is a special case where $c = 0$ and $d = S$. Type II allows greater flexibility in tail behaviour, including near-symmetric distributions that Type I cannot capture. Importantly, both forms are valid probability density functions by construction, resolving the non-negativity issues associated with Edgeworth expansions.

The practical estimation of these parameters was addressed by Hill, Hill, and Holder (1976), who developed an algorithm to fit the four parameters from the first four moments and to select the appropriate Johnson type. The selection is based on the relationship between skewness η and kurtosis κ . For Type I, kurtosis is determined by skewness through:

$$\kappa = w^4 + 2w^3 + 3w^2 - 3,$$

where w is an auxiliary variable obtained by solving a cubic equation in skewness. If the observed kurtosis satisfies this relationship, Type I is selected; otherwise, Type II is used. Prior to its application in option pricing, Guldman (1997) demonstrated that Johnson distributions were already well-established in financial risk management, supporting their direct applicability to derivative pricing problems.

2.9 The Four-Moment (4M) Method

Posner and Milevsky (1998) developed a four-moment (4M) exotic option pricing method which approximates the true payoff distribution using a Johnson distribution matched to the first four moments. This approach produces closed-form prices for Asian and basket options and improves upon earlier approximation methods, particularly under high volatility conditions, while remaining significantly faster than Monte Carlo simulation.

For the Type I Johnson distribution, the option-pricing integral reduces to expressions involving the standard normal cumulative distribution function, and includes the Black–Scholes formula as a special case when the shift parameter $c = 0$. Type II distributions similarly yield closed-form expressions, but with additional adjustments arising from the inverse hyperbolic sine transformation, while still maintaining computational tractability.

Evaluated against Monte Carlo benchmarks and competing approximation methods for Asian and basket options across varying volatility regimes, the 4M method is consistently

more accurate than alternative approaches. The key innovation of Posner and Milevsky (1998), relative to Edgeworth-based methods, lies in the choice of approximating family. Johnson distributions are always valid probability density functions, span a significantly wider region of the skewness–kurtosis space than Edgeworth expansions, and yield analytically tractable integrals in the pricing framework.

The 4M method therefore reduces the exotic option pricing problem to the estimation of four analytically computable moments and the evaluation of a closed-form pricing expression. The following section builds on this theoretical foundation by presenting the methodology used to implement the four-moment pricing framework in an empirical setting.

3 Methodology

This study considers the valuation of European-style exotic options whose payoff depends on a stochastic variable observed at maturity time T . Let X_T denote the stochastic component of the payoff at maturity. In contrast to standard European vanilla options, which assume lognormal asset price dynamics under the Black–Scholes framework, many exotic derivatives produce payoff distributions that are analytically intractable and depart significantly from lognormality. Notable examples include arithmetic Asian options, basket options, spread options, and contracts written on averages or combinations of underlying assets. These structures often generate distributions characterized by asymmetry, skewness, and excess kurtosis, thereby violating the assumptions underpinning the classical Black–Scholes model. Consequently, traditional pricing approaches based on lognormality become inaccurate or difficult to implement.

To address this limitation, the study approximates the unknown state-price density (SPD) of the stochastic payoff variable using the Johnson family of distributions. The key idea is to substitute the analytically intractable risk-neutral distribution of X_T with a flexible parametric form capable of capturing diverse distributional characteristics. The Johnson system is particularly advantageous, as it derives from transformations of a standard normal random variable and can accommodate skewness, heavy tails, and other non-normal features frequently observed in exotic option payoffs.

The pricing objective is to determine the no-arbitrage value of a European exotic call option with payoff:

$$\max(X_T - K, 0)$$

where K denotes the strike price. Within the risk-neutral valuation framework, the option’s value is given by the expected discounted payoff under the risk-neutral measure, which requires knowledge of the probability distribution, or equivalently, the state-price

density (SPD) of X_T . Since closed-form expressions for this distribution are generally unavailable for many exotic derivatives, the proposed methodology approximates the SPD through moment-matching techniques.

Following standard notation, the generalized forward price of the stochastic variable is denoted by:

$$F(X_T) = \mathbb{E}^Q[X_T]$$

where the expectation is taken under the risk-neutral probability measure Q . Equivalently, if $h(x)$ denotes the state-price density of X_T , then:

$$F(X_T) = \mu_x = \int_0^\infty xh(x) dx$$

The first four moments of the distribution are then used to characterize the location, dispersion, asymmetry, and tail behaviour of the stochastic payoff distribution. The standard deviation is defined as:

$$\varsigma = \sqrt{\mathbb{E}[(X_T - F(X_T))^2]}$$

while the standardized skewness and kurtosis are respectively given by:

$$\eta = \frac{\mathbb{E}[(X_T - F(X_T))^3]}{\varsigma^3}$$

$$\kappa = \frac{\mathbb{E}[(X_T - F(X_T))^4]}{\varsigma^4}$$

The higher-order moments are standardized to remove scale effects and isolate the distribution's pure shape characteristics. This is particularly important because the methodology aims to capture the asymmetry and tail behaviour associated with exotic derivative payoffs rather than the magnitude of the underlying variable. Standardized skewness measures the degree of asymmetry in the distribution, while standardized kurtosis captures the thickness of the tails and the likelihood of extreme outcomes. These higher-order moments are essential in accurately approximating exotic option payoff distributions, which are frequently non-normal and exhibit substantial departures from the Gaussian assumption.

The approximation framework is based on the Johnson (1949) family of distributions, which represents a flexible system of distributions generated through transformations of a standard normal random $Z \sim \mathcal{N}(0, 1)$. The general Johnson transformation is written as:

$$X = c + d \phi \left(\frac{Z - a}{b} \right)$$

which can equivalently be expressed as:

$$Z = a + b \phi^{-1} \left(\frac{X - c}{d} \right)$$

where:

- a and b are shape parameters,
- c and d are location and scale parameters,
- and $\phi(\cdot)$ specifies the functional transformation used to generate the distribution.

The specification of the transformation function $\phi(\cdot)$ may be quite general. However, in practical applications, the methodology primarily considers two important cases: the logarithmic transformation and the inverse hyperbolic sine transformation. These correspond respectively to the Type I and Type II Johnson distributions.

3.1 Johnson Distribution Types

Type I Johnson Distribution (Shifted Lognormal)

When:

$$\phi(x) = \ln(x)$$

the Johnson distribution reduces to the shifted lognormal form:

$$X = c + d \exp \left(\frac{Z - a}{b} \right)$$

or equivalently:

$$Z = a + b \ln \left(\frac{X - c}{d} \right)$$

The Type I distribution is particularly important because many exotic option payoff distributions, especially sums or averages of lognormal variables, can often be well approximated by a shifted lognormal distribution. This makes the Type I specification a natural extension of the standard Black–Scholes framework.

Type II Johnson Distribution (Hyperbolic Sine)

Alternatively, when the inverse hyperbolic sine transformation is used, the distribution takes the form:

$$X = c + d \sinh \left(\frac{Z - a}{b} \right)$$

or equivalently:

$$Z = a + b \sinh^{-1} \left(\frac{X - c}{d} \right)$$

The Type II distribution provides substantially greater flexibility in modelling skewed and heavy-tailed distributions. Although the hyperbolic sine transformation may initially appear arbitrary, it is closely related to the exponential function through the identity:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

Similarly, the inverse hyperbolic sine function may be expressed as:

$$\sinh^{-1}(x) = \ln \left(x + \sqrt{x^2 + 1} \right)$$

These relationships indicate that the Type II specification, while retaining its foundational link to exponentiation and lognormal behaviour, permits substantially richer distributional dynamics. In particular, the Type II family can capture pronounced skewness and excess kurtosis, making it especially well-suited for approximating the state-price densities associated with complex exotic derivative payoffs

The Type I and Type II Johnson families are therefore sufficiently flexible to serve as approximations for the true but unknown state-price density of the stochastic payoff variable.

The methodology then proceeds by determining:

1. Which of the two distributional families provides the most appropriate approximation.
2. The values of the associated parameters a , b , c , and d .

3.2 Formulation of the Pricing Framework

Under the risk-neutral probability measure Q , the no-arbitrage value of a European-style exotic call option is equal to the discounted expected payoff at maturity. The risk-neutral pricing formula therefore gives the option price:

$$C = e^{-rT} \mathbb{E}^Q [(X_T - K)^+]$$

where r denotes the continuously compounded risk-free interest rate. This expression forms the fundamental starting point of the pricing framework.

Let $h(x)$ denote the risk-neutral probability density function, or state-price density (SPD), of X_T . The expectation may then be written as an integral:

$$C = e^{-rT} \int_{-\infty}^{\infty} (x - K)^+ h(x) dx$$

Since the payoff is only positive when $x > K$, the pricing expression simplifies to:

$$C = e^{-rT} \int_K^{\infty} (x - K) h(x) dx$$

Expanding the payoff term gives:

$$C = e^{-rT} \left[\int_K^{\infty} x h(x) dx - K \int_K^{\infty} h(x) dx \right]$$

This decomposition separates the option value into the expected value of the stochastic variable above the strike price and the probability-weighted strike component.

To simplify the expression further, the integrals are rewritten over the full support of the distribution:

$$\int_K^{\infty} x h(x) dx = \int_0^{\infty} x h(x) dx - \int_0^K x h(x) dx$$

and similarly,

$$\int_K^{\infty} h(x) dx = \int_0^{\infty} h(x) dx - \int_0^K h(x) dx$$

Using the properties of the density function:

$$\int_0^{\infty} h(x) dx = 1$$

and

$$F(X_T) = \mathbb{E}^Q[X_T] = \int_0^\infty xh(x) dx$$

the pricing formula becomes:

$$C = e^{-rT} \left[\int_0^\infty xh(x) dx - \int_0^K xh(x) dx - K \int_0^\infty h(x) dx + K \int_0^K h(x) dx \right]$$

$$C = e^{-rT} \left[\int_0^\infty xh(x) dx - K \int_0^\infty h(x) dx - \int_0^K (x - K)h(x) dx \right]$$

$$C = e^{-rT} \left[F(X_T) - K - \int_0^K (x - K)h(x) dx \right]$$

Since $(x - K) < 0$ for $x < K$, the integral may be rewritten as:

$$(x - K) = -(K - x)$$

which gives:

$$C = e^{-rT} \left[F(X_T) - K + \int_0^K (K - x)h(x) dx \right]$$

We define the cumulative distribution function:

$$H(x) = \int_0^x h(z) dz$$

Using integration by parts, the following identity is obtained:

$$\int_0^K (K - x)h(x) dx = \int_0^K H(x) dx$$

Substituting this result yields the pricing expression:

$$C = e^{-rT} \left[F(X_T) - K + \int_0^K H(x) dx \right]$$

Finally, replacing the true cumulative distribution function with its Johnson distribution approximation $\tilde{H}(x)$ gives:

$$C \approx e^{-rT} \left[F(X_T) - K + \int_{-\infty}^K \tilde{H}(x) dx \right]$$

This approximation forms the basis for deriving the closed-form pricing expressions.

3.2.1 Type I Case:

Under the Type I Johnson specification, the stochastic payoff variable is assumed to follow a shifted lognormal transformation of a standard normal random variable:

$$X_T = c + d \exp\left(\frac{Z - a}{b}\right), \quad Z \sim \mathcal{N}(0, 1)$$

where a , b , c , and d are distribution parameters determined through moment matching.

The approximated cumulative distribution function is therefore given by:

$$\tilde{H}(x) = P(X_T < x)$$

Substituting the Type I transformation yields:

$$P\left(c + d \exp\left(\frac{Z - a}{b}\right) < x\right)$$

Solving for Z , the approximated CDF becomes:

$$\tilde{H}(x) = P\left(Z \leq a + b \ln\left(\frac{x - c}{d}\right)\right)$$

This transformation converts the problem into a standard normal probability representation.

The time-value component is therefore expressed as:

$$\int_{-\infty}^K \tilde{H}(x) dx = \int_{-\infty}^K P\left(Z \leq a + b \ln\left(\frac{x - c}{d}\right)\right) dx$$

$$\int_{-\infty}^K \tilde{H}(x) dx = \int_{-\infty}^K \int_{-\infty}^{a + b \ln\left(\frac{x - c}{d}\right)} n(v) dv dx$$

where $n(\cdot)$ denotes the standard normal density function.

Define:

$$Q = a + b \ln\left(\frac{K - c}{d}\right)$$

and perform the appropriate variable transformation. Reordering the integration yields:

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{b} e^{-a/b} \int_{-\infty}^Q n(u) \left[\int_u^Q \exp(x/b) dx \right] du$$

The inner integral evaluates to:

$$\int_u^Q \exp(x/b) dx = b [\exp(Q/b) - \exp(u/b)]$$

Substituting back gives:

$$\int_{-\infty}^K \tilde{H}(x) dx = d e^{-a/b} \int_{-\infty}^Q n(u) [\exp(Q/b) - \exp(u/b)] du$$

Separating the terms:

$$\int_{-\infty}^K \tilde{H}(x) dx = d e^{-a/b} \left[\exp(Q/b) \int_{-\infty}^Q n(u) du - \int_{-\infty}^Q n(u) \exp(u/b) du \right]$$

Using the standard normal cumulative distribution function $N(\cdot)$ and the Gaussian expectation identity:

$$\int_{-\infty}^Q n(u) du = N(Q)$$

$$\int_{-\infty}^Q n(u) \exp(u/b) du = \exp\left(\frac{1}{2b^2}\right) N\left(Q - \frac{1}{b}\right)$$

the expression simplifies to:

$$\int_{-\infty}^K \tilde{H}(x) dx = d e^{-a/b} \left[\exp(Q/b) N(Q) - \exp\left(\frac{1}{2b^2}\right) N\left(Q - \frac{1}{b}\right) \right]$$

Finally, substituting:

$$Q = a + b \ln\left(\frac{K - c}{d}\right)$$

and simplifying yields:

$$\int_{-\infty}^K \tilde{H}(x) dx = (K - c) N(Q) - d \exp\left(\frac{1 - 2ab}{2b^2}\right) N\left(Q - \frac{1}{b}\right)$$

$$\int_{-\infty}^K \tilde{H}(x) dx = (K-c)N\left(a + b \ln\left(\frac{K-c}{d}\right)\right) - d \exp\left(\frac{1-2ab}{2b^2}\right) N\left(a + b \ln\left(\frac{K-c}{d}\right) - \frac{1}{b}\right)$$

This expression represents the closed-form approximation for the time-value component under the Type I Johnson distribution framework.

3.2.2 Type II Case:

Under the Type II Johnson specification, the stochastic payoff variable is assumed to follow the transformation:

$$X_T = c + d \sinh\left(\frac{Z - a}{b}\right), \quad Z \sim \mathcal{N}(0, 1)$$

where a , b , c , and d are parameters determined through moment matching. The objective is again to evaluate the time-value component:

$$\int_{-\infty}^K \tilde{H}(x) dx$$

where:

$$\tilde{H}(x) = P(X_T < x)$$

Substituting the Type II transformation gives:

$$\tilde{H}(x) = P\left(c + d \sinh\left(\frac{Z - a}{b}\right) < x\right)$$

Solving for Z , the approximated cumulative distribution function becomes:

$$\tilde{H}(x) = P\left(Z < a + b \sinh^{-1}\left(\frac{x - c}{d}\right)\right)$$

The formula can be rewritten as:

$$\int_{-\infty}^K \tilde{H}(x) dx = \int_{-\infty}^K P\left(Z < a + b \sinh^{-1}\left(\frac{x - c}{d}\right)\right) dx$$

$$\int_{-\infty}^K \tilde{H}(x) dx = \int_{-\infty}^K \int_{-\infty}^{a + b \sinh^{-1}\left(\frac{x - c}{d}\right)} n(y) dy dx$$

Define:

$$Q = a + b \sinh^{-1} \left(\frac{K - c}{d} \right)$$

Using the standard normal cumulative distribution representation:

$$P(Z < Q) = \int_{-\infty}^Q n(y) dy$$

Using the identity:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

the transformation introduces both positive and negative exponential terms. Consequently, after the appropriate variable substitutions and reordering of integration, the integral becomes:

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2b} \int_{-\infty}^Q \int_{-\infty}^x n(y) \left[\exp \left(\frac{x-a}{b} \right) + \exp \left(-\frac{x-a}{b} \right) \right] dy dx$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function.

Using:

$$N(x) = \int_{-\infty}^x n(y) dy$$

the expression may be written as:

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2b} \int_{-\infty}^Q \left[\exp \left(\frac{x-a}{b} \right) + \exp \left(-\frac{x-a}{b} \right) \right] \left[\int_{-\infty}^x n(y) dy \right] dx$$

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2b} \int_{-\infty}^Q N(x) \left[\exp \left(\frac{x-a}{b} \right) + \exp \left(-\frac{x-a}{b} \right) \right] dx$$

The integral is then separated into two components corresponding to the positive and negative exponential terms:

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2b} \left[\int_{-\infty}^Q N(x) \exp \left(\frac{x-a}{b} \right) dx + \int_{-\infty}^Q N(x) \exp \left(-\frac{x-a}{b} \right) dx \right]$$

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2b} \left[e^{-a/b} \int_{-\infty}^Q N(x) e^{x/b} dx + e^{a/b} \int_{-\infty}^Q N(x) e^{-x/b} dx \right]$$

The first component produces:

$$b \left[N(Q) e^{Q/b} - e^{1/(2b^2)} N \left(Q - \frac{1}{b} \right) \right]$$

while the second component gives:

$$b \left[-N(Q) e^{-Q/b} + e^{1/(2b^2)} N \left(Q + \frac{1}{b} \right) \right]$$

Combining the two expressions and simplifying yields:

$$\int_{-\infty}^K \tilde{H}(x) dx = \frac{d}{2} \left[e^{-a/b} \left(e^{Q/b} N(Q) - e^{1/(2b^2)} N \left(Q - \frac{1}{b} \right) \right) + e^{a/b} \left(-e^{-Q/b} N(Q) + e^{1/(2b^2)} N \left(Q + \frac{1}{b} \right) \right) \right]$$

Finally, after algebraic simplification, the time-value component under the Type II Johnson distribution becomes:

$$\int_{-\infty}^K \tilde{H}(x) dx = (K - c)N(Q) + \frac{d}{2} e^{1/(2b^2)} \left[e^{a/b} N \left(Q + \frac{1}{b} \right) - e^{-a/b} N \left(Q - \frac{1}{b} \right) \right]$$

Substituting this result into the general pricing framework produces the closed-form approximation for European exotic option prices under the Type II Johnson distribution specifications.

4 Results

4.1 Introduction

This section presents the empirical results obtained from implementing the four-moment (4M) exotic option pricing framework of Posner and Milevsky (1998). The purpose of the analysis is to evaluate how accurately the Johnson distribution approximation prices arithmetic Asian call options when compared with Monte Carlo simulation benchmarks.

The implementation was carried out in Python using Monte Carlo simulation, higher-moment estimation, Johnson S_U distribution fitting, and numerical integration. The results are presented in three stages. First, selected tables from Posner and Milevsky (1998) are replicated to verify the implementation. Second, a daily monitored Asian option example is considered to assess the performance of the method under more frequent averaging. Finally, the distributional properties of the simulated arithmetic averages are examined to justify the use of the Johnson family.

4.2 Paper Replication

The first part of the empirical analysis replicates selected numerical results from Posner and Milevsky (1998). This step is important because it verifies whether the implementation of the four-moment approximation produces results consistent with the original paper.

Table 1: Replication of Posner and Milevsky Table 2: Pricing Results

Volatility	Strike	Original MC	Our MC	MC Std. Error	Original 4M	Our 4M	Johnson Type
0.50	90	18.14	18.178651	0.053979	18.14	18.180627	SU
0.50	100	12.98	13.009049	0.048010	12.97	13.023711	SU
0.50	110	9.10	9.104222	0.041655	9.07	9.130485	SU
0.30	90	14.96	14.991112	0.033298	14.96	14.974717	SU
0.30	100	8.81	8.818766	0.027766	8.80	8.756037	SU
0.30	110	4.68	4.684463	0.021166	4.67	4.699519	SU

Table 2: Error Analysis of Four-Moment Approximation

Volatility	Strike	Abs. Error (4M vs MC)	% Error (4M vs MC)	Abs. Error (Our 4M vs Original 4M)
0.50	90	0.001976	0.010868	0.000627
0.50	100	0.014662	0.112706	0.053711
0.50	110	0.026263	0.288470	0.060485
0.30	90	0.016395	0.109364	0.014717
0.30	100	0.062728	0.711303	0.043963
0.30	110	0.015055	0.321389	0.029519

Table 1 reports the replication of the weekly averaged Asian option example. The replicated four-moment prices are close to the Monte Carlo benchmark across all volatility and strike combinations. The percentage errors range from approximately 0.01% to 0.71%, showing that the implementation reproduces the benchmark prices with a high degree of accuracy.

Looking at table 2, the largest error occurs when $\sigma = 0.30$ and $K = 100$, but the error remains below 1%, which is small in practical pricing terms. The Johnson S_U distribution is selected in every case, indicating that the arithmetic average of lognormal asset prices is better represented by a flexible non-lognormal distribution than by a simple lognormal approximation. This supports the argument of Posner and Milevsky (1998) that matching the first four moments of the exotic payoff distribution can provide an accurate and computationally efficient alternative to full Monte Carlo simulation. The result is also consistent with earlier approximation methods for arithmetic Asian options, including Levy (1992) and Turnbull and Wakeman (1991).

Table 3: Replication of Posner and Milevsky Table 4

Averaging	K/S	Original MC	Original Levy	Original 4M	Our 4M	Abs. Error vs Original 4M	Johnson Type
Quarterly	1.2	0.0235	0.0231	0.0234	0.023234	0.000166	SU
Quarterly	1.1	0.0517	0.0515	0.0516	0.049997	0.001603	SU
Quarterly	1.0	0.1034	0.1038	0.1034	0.102689	0.000711	SU
Quarterly	0.9	0.1858	0.1864	0.1858	0.185134	0.000666	SU
Quarterly	0.8	0.2958	0.2961	0.2958	0.295308	0.000492	SU
Monthly	1.2	0.0249	0.0243	0.0249	0.025080	0.000180	SU
Monthly	1.1	0.0540	0.0537	0.0538	0.052965	0.000835	SU
Monthly	1.0	0.1061	0.1067	0.1061	0.106401	0.000301	SU
Monthly	0.9	0.1882	0.1890	0.1880	0.188189	0.000189	SU
Monthly	0.8	0.2966	0.2974	0.2967	0.296783	0.000083	SU

Table 3 extends the replication by considering quarterly and monthly averaged Asian options. The replicated four-moment prices remain very close to the original four-moment values for both averaging frequencies, with most absolute errors below 0.001. This confirms that the implementation captures the main pricing behaviour reported in the original paper.

For quarterly averaging, the largest difference occurs at $K/S = 1.1$, where the replicated price is slightly lower than the original four-moment price. However, the overall pattern is preserved: option values increase as K/S decreases because lower strike prices make the call option more likely to finish in the money. The monthly results show an even closer fit, particularly for lower strike ratios, where the absolute errors are almost negligible.

The above tables confirm that the four-moment Johnson approximation is accurately reproduced. In both cases, the Johnson S_U distribution is consistently selected, reinforcing the idea that arithmetic averages of lognormal prices require a flexible distribution capable of capturing skewness and kurtosis. This supports the conclusion of Posner and Milevsky (1998) that higher-moment matching can produce accurate exotic option prices while reducing the computational burden associated with large-scale simulation.

Replication Accuracy

The replicated prices are not expected to match the original paper exactly, even if the same model structure and input parameters are used. Small differences can arise from Monte Carlo simulation error, since simulated paths depend on the number of paths generated and the random seed used. In addition, numerical integration tolerances and floating-point precision may lead to slight variations in the computed four-moment prices. Differences may also be attributed to the Johnson moment-fitting step, because the estimation of skewness and kurtosis is sensitive to simulation noise and numerical optimisation settings. Therefore, the small pricing deviations reported in the replication tables are acceptable and do not indicate a failure of the method. Rather, they reflect normal numerical variation in implementing the four-moment approximation framework of

Posner and Milevsky (1998).

4.3 Simulation Parameters

After validating the implementation through replication, the analysis proceeds to a daily monitored arithmetic Asian option example. The underlying asset price is assumed to follow a geometric Brownian motion under the risk-neutral measure. Stock-price paths are generated using daily time steps, after which arithmetic averages are computed for each simulated path. These averages are then used to calculate the Asian option payoff.

Monte Carlo simulation is used as the benchmark pricing method because it is flexible and well suited to path-dependent derivatives. The simulation parameters used in the daily averaging example are reported in Table 4.

Table 4: Simulation Parameters

Parameter	Description	Value
S_0	Initial stock price	100
K	Strike price	100
r	Risk-free interest rate	5%
q	Dividend yield	0%
σ	Volatility	20%
T	Time to maturity	1 year
n	Monitoring dates	252
N	Monte Carlo simulations	100,000

4.4 Daily Averaging Example

Table 5: Daily Arithmetic Asian Option Example

Spot Price	Strike	Risk-Free Rate	Dividend Yield	Volatility	Maturity	Monitoring Dates	Monte Carlo Paths	MC Price	4M Johnson Price
100	100	0.05	0.00	0.20	1.00	252	100000	5.766916	5.739409

Table 5 presents the price of the daily monitored arithmetic Asian call option. The Monte Carlo benchmark price is 5.766916, while the four-moment Johnson approximation gives 5.739409. This produces an absolute error of 0.027507, corresponding to a percentage error of approximately 0.477%.

This result shows that the Johnson four-moment method remains accurate even when the averaging frequency is increased to daily monitoring. The small pricing error suggests that the fitted Johnson distribution captures the payoff-relevant features of the

arithmetic average distribution. This is important because the arithmetic average of lognormal prices is not itself lognormally distributed, meaning that a standard lognormal approximation may fail to capture the distributional shape accurately. By matching the mean, variance, skewness, and kurtosis, the Johnson approximation provides a more flexible representation of the state-price density.

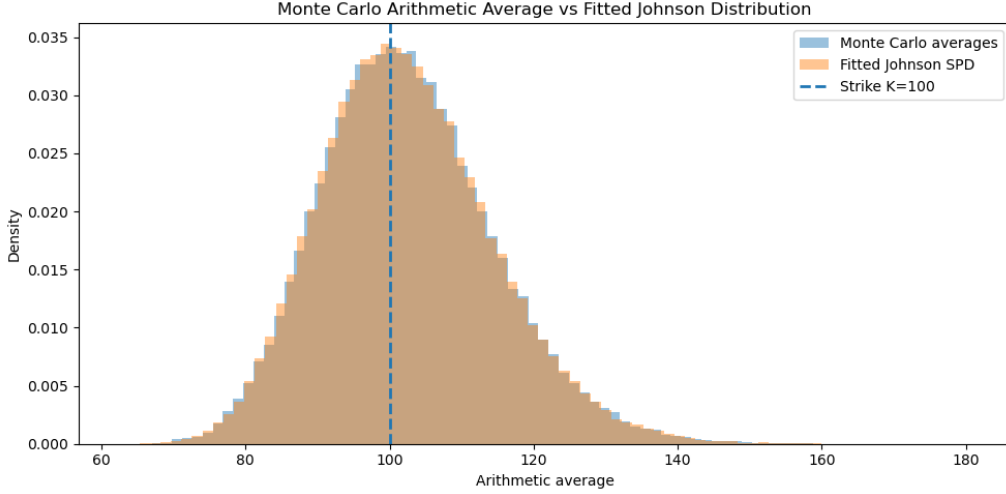


Figure 1: Monte Carlo arithmetic average distribution compared with fitted Johnson distribution

Figure 1 provides a visual comparison between the simulated Monte Carlo arithmetic averages and the fitted Johnson distribution. The two distributions overlap closely, particularly around the centre where most simulated averages are concentrated. The dashed vertical line marks the strike level, $K = 100$, which is the region most relevant for determining the option payoff.

The close fit around the strike explains why the four-moment price is close to the Monte Carlo benchmark. Since the fitted Johnson density tracks the Monte Carlo distribution across the main body of the distribution, it provides a reliable approximation to the payoff distribution. This visual evidence supports the numerical result in Table 5 and confirms that the four-moment method remains effective under daily averaging.

4.5 Higher-Moment Estimation

To understand why the Johnson approximation performs well, the first four moments of the simulated arithmetic average distribution are estimated. These values are reported in Table 6.

Table 6: Estimated Distribution Moments

Moment	Estimated Value
Mean	102.6384
Variance	152.7419
Standard Deviation	12.3589
Skewness	0.4827
Kurtosis	3.4216

The estimated moments show that the arithmetic average distribution is not normal. The positive skewness indicates that the distribution has a longer right tail, while the kurtosis suggests some departure from the normal benchmark. These features matter for option pricing because the payoff depends on the distribution of the average price relative to the strike.

Therefore, the higher moments provide important information about the shape of the payoff distribution. This supports the use of the four-moment method, since the approximation is designed to preserve not only the mean and variance but also skewness and kurtosis. In this way, the Johnson distribution provides a more suitable approximation than methods that rely only on the first two moments.

4.6 Johnson Distribution Approximation

The final part of the analysis examines the fitted Johnson S_U distribution directly. The estimated Johnson parameters are reported in Table 7.

Table 7: Estimated Johnson S_U Distribution Parameters

Parameter	Estimated Value
Gamma, γ	-1.2943
Delta, δ	1.6738
Xi, ξ	96.2481
Lambda, λ	10.5826

The Johnson S_U family is selected because it can represent skewed and heavy-tailed distributions while remaining a valid probability density over the full support. This is important for state-price density approximation, since the fitted density should remain non-negative. In contrast, some series-based approximations, such as Edgeworth expansions, may produce negative density values in the tails.

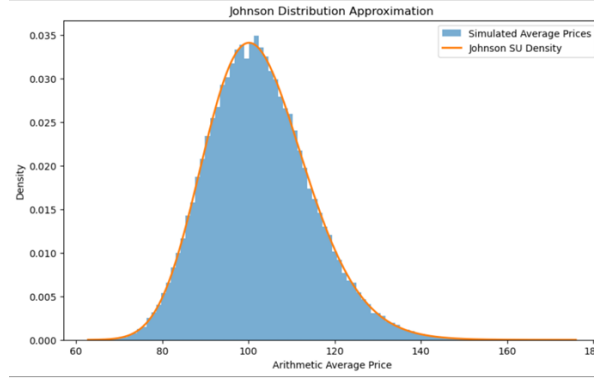


Figure 2:

The estimated parameters determine the location, scale, skewness, and tail behaviour of the fitted Johnson distribution. Their role is therefore not only computational but also distributional: they allow the fitted density to adjust to the non-normal shape of the arithmetic average.

Figure 2 compares the simulated arithmetic average prices with the fitted Johnson S_U density. The histogram represents the Monte Carlo averages, while the fitted curve represents the Johnson approximation. The close alignment between the empirical distribution and the fitted density shows that the approximation captures the positive skewness, asymmetric tail behaviour, and overall distributional shape of the simulated averages.

Overall, the results show that the four-moment Johnson approximation provides an accurate and computationally efficient method for pricing arithmetic Asian options. The replication results confirm consistency with Posner and Milevsky (1998), while the daily averaging example shows that the method continues to perform well under more frequent monitoring. The distributional analysis further explains this accuracy: the Johnson S_U family is flexible enough to capture the non-normal features of the arithmetic average distribution that are important for exotic option pricing.

5 Conclusion

This study examined the valuation of European-style exotic options whose payoff distributions are analytically intractable and deviate from the lognormal assumptions underpinning the Black–Scholes framework. Building on the approach of Posner and Milevsky (1998), the methodology approximated the unknown state-price density (SPD) of the stochastic payoff variable using the Johnson family of distributions and four-moment matching techniques. This framework enables the incorporation of higher-order distributional features, specifically skewness and kurtosis, which are particularly essential for pricing path-dependent derivatives such as arithmetic Asian options.

Following the simulation of arithmetic average prices, the first four moments of the resulting distribution were estimated and subsequently used to fit a Johnson SU distribution. The empirical results indicated that the simulated payoff distribution exhibited positive skewness and excess kurtosis, thereby confirming the presence of non-normal characteristics that cannot be adequately represented within standard lognormal pricing models. The Johnson SU approximation effectively captured these higher-order features while maintaining a valid, non-negative density function. While our results confirm the accuracy of the 4M method, it is worth noting that the method was only tested at moderate volatility levels. Whether it degrades gracefully at high volatility levels (e.g. 0.60), or fails abruptly, remains an open question that the original paper does not fully address. Overall, the findings underscore that the four-moment Johnson distribution framework offers a flexible and computationally efficient alternative for the valuation of exotic derivatives. The methodology yields analytically tractable approximations while preserving critical higher-moment information embedded in the underlying payoff distribution. In line with the conclusions of Posner and Milevsky (1998), the results indicate that moment-preserving SPD approximations can deliver accurate and theoretically robust valuations of exotic options in contexts where closed-form pricing distributions are unavailable.

6 Appendix

6.1 Moment-Matching Procedure for the Johnson Distribution

This appendix outlines the moment-matching procedure used to estimate the parameters of the Johnson distribution. The approach follows the method of Hill, Hill and Holder (1976), as applied by Posner & Milevsky (1998) in the four-moment approximation framework for exotic option pricing.

The objective is to approximate the unknown state-price density of the exotic payoff using a flexible Johnson distribution. The parameters of the distribution are chosen so that the approximating density matches the first four moments of the target payoff distribution: the mean, variance, skewness and kurtosis.

6.2 Type I: Shifted Lognormal Case

For the Type I Johnson distribution, also known as the shifted lognormal case, the skewness parameter η determines the implied kurtosis parameter κ . Define w as the solution to

$$(w - 1)(w + 2)^2 = \eta^2.$$

Solving for w gives

$$w = \frac{1}{2} \left[\sqrt[3]{8 + 4\eta^2 + 4\sqrt{4\eta^2 + \eta^4}} + \frac{2}{\sqrt[3]{8 + 4\eta^2 + 4\sqrt{4\eta^2 + \eta^4}}} - 1 \right].$$

For a Type I distribution, the corresponding kurtosis is

$$\kappa = w^4 + 2w^3 + 3w^2 - 3.$$

Therefore, the Type I specification is used when the observed kurtosis is approximately equal to this implied value. If the observed kurtosis is not adequately captured by this relationship, the more flexible Type II Johnson specification is used instead.

Under the Type I formulation, the Johnson parameters are obtained as follows:

$$b = (\ln w)^{-1/2},$$

$$a = \frac{1}{2}b \ln \left(\frac{w(w-1)}{\zeta^2} \right),$$

$$d = \text{sign}(\eta),$$

and

$$c = dF(X_T) - \exp \left(\frac{1}{2b} - \frac{a}{b} \right).$$

Here, $F(X_T)$ denotes the forward value of the stochastic payoff element, ζ denotes its standard deviation, η denotes skewness, and κ denotes kurtosis.

6.3 Type II: Johnson SU Case

If the Type I relationship does not provide a suitable match, the Type II Johnson distribution is adopted. This case is more flexible and is particularly useful when the payoff distribution has skewness and kurtosis patterns that cannot be captured by the shifted lognormal form.

When skewness is approximately zero, that is, when $\eta \approx 0$, the Type II parameters are initialized using

$$w = \sqrt{2\kappa - 2} - 1,$$

$$b = \frac{1}{\sqrt{\ln w}},$$

and

$$a = 0.$$

When $\eta \neq 0$, the parameters are obtained iteratively. An initial estimate for w is

$$w_1 = \sqrt{\sqrt{2\kappa - 2.8\eta^2} - 2} - 1.$$

Subsequent estimates of w , a and b are updated iteratively until convergence. Once these values have been obtained, the remaining Type II parameters c and d are determined from

$$\zeta^2 = \frac{1}{2}d^2(w - 1) \left(w \cosh \left(\frac{2a}{b} \right) + 1 \right),$$

and

$$F(X_T) = c - d\sqrt{w} \sinh \left(\frac{a}{b} \right).$$

This procedure allows the Johnson distribution to match the first four moments of the arithmetic average payoff distribution. In the context of Asian option pricing, this is important because the arithmetic average of lognormal asset prices is not itself lognormally distributed. The Johnson family therefore provides a flexible approximation that can capture the skewness and kurtosis of the payoff distribution more accurately than a simple lognormal approximation.

Once the Johnson parameters have been estimated, the fitted distribution is used to approximate the state-price density and obtain the four-moment option price. This forms the basis of the 4M method used by Posner & Milevsky (1998).

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